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Editors
Journal of the American Statistical Association

Dear Editors,

re: “Quantifying Moderator Effects on High-Dimensional Multi-Subject Time Series” by José Á. Sánchez Gómez, Holly O’Rourke and Gene Brewer.

Could you please consider the above article for possible publication in *JASA*?

Researchers across many areas use time-series data to estimate relationships among variables over time. These relationships can be modeled via Granger-causal networks (GCNs), where nodes correspond to variables and directed edges indicate lagged temporal relationships among them. When time-series data are gathered over multiple subjects, GCN models must account for the presence of shared and subject-specific edges across the GCNs. While existing methods can estimate GCNs on multi-subject data, they often lack interpretable explanations for the source of the observed heterogeneity. In the present paper, we propose the moderator vector autoregressive model (MOD-VAR) for interpretable estimation of GCNs in multi-subject time-series data. To this end, we decompose the GCNs into shared, moderator-dependent, and subject-specific components. The use of moderator-dependent components can enhance both the quantification and interpretability of heterogeneous GCNs. Our theoretical results establish that MOD-VAR improves multi-subject GCN estimation in high-dimensional regimes with substantial heterogeneity. Furthermore, extensive numerical simulations demonstrate that MOD-VAR outperforms existing methods in terms of both GCN estimation and forecasting in a variety of settings.

Thank you very much for your consideration. I look forward to hearing from you.

Yours sincerely

José Á. Sánchez Gómez